Scale and renormalization-group audit of *Clifford Geometry and Mass Relations*, v13

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Abstract

The manuscript uses the Thomson-limit fine-structure constant with physical lepton pole masses. We test whether the lepton and proton relations prefer a different electromagnetic momentum scale, and run charged-lepton masses to a common $\overline{\mathrm{MS}}$ scale at one loop. The three formulas require mutually different effective momenta when pole masses are held fixed. The internally motivated geometric momenta, $1/r_e$ and E_*, worsen the agreement. A consistent running-mass comparison produces no intersection at a common scale above m_τ; the Koide relation is closer to 2/3 for pole than for running masses. These are one-loop tests, not a complete electroweak/QCD precision calculation.

1. What is being evolved?

The manuscript states three *pole-mass* relations in terms of $x = \alpha^{-1}(0)$:

$$R_\mu = \frac{3}{2}x + \frac{17}{14}, \qquad R_\tau = \frac{51}{2}x - 17, \qquad R_p = \frac{27}{2}x - \frac{83}{6} + \frac{1}{3x(x-1)}. \tag{1}$$

The fixed inputs are $x_0 = 137.035999177$, $m_e c^2 = 0.51099895069$ MeV, $R_\mu^{\mathrm{obs}} = 206.7682827$, $m_\tau c^2 = 1776.93$ MeV, and $R_p^{\mathrm{obs}} = 1836.152673426$. The tau figure is the PDG comparison employed in the earlier audit. We compare two distinct operations: (i) substituting a spacelike effective charge while keeping pole masses fixed, which is only a phenomenological diagnostic, and (ii) consistently evolving running masses and the coupling in a common renormalization scheme.

2. Phenomenological substitution of the spacelike effective charge

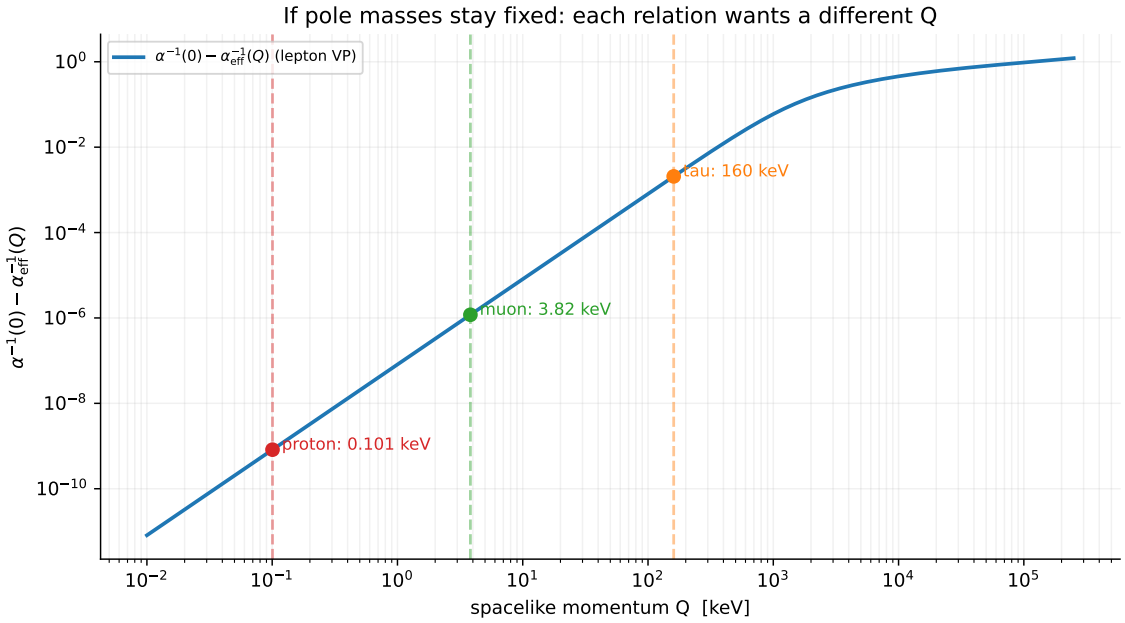
At one-loop with lepton vacuum polarization only, the momentum-subtracted inverse effective coupling is

$$x_{\mathrm{eff}}(Q) = x_0 - \frac{2}{\pi}\sum_{\ell=e,\mu,\tau}\int_0^1 dt\, t(1-t)\ln\left[1 + \frac{Q^2 t(1-t)}{m_\ell^2}\right], \qquad Q^2 > 0 \text{ (spacelike)}. \tag{2}$$

The Q values in the next table are the separate solutions of $R_j(x_{\mathrm{eff}}(Q)) = R_j^{\mathrm{obs}}$. At these small scales, lepton vacuum polarization dominates and hadronic corrections are negligible.

Target formula	Required $x = \alpha^{-1}$	$x_0 - x$	Inferred Q
Proton	137.035999176178	8.22×10^{-10}	0.1006 keV
Muon	137.035997990476	1.187×10^{-6}	3.821 keV
Tau	137.033931839465	0.00206734	160.33 keV

The proton shift in x is substantially below the stated input standard uncertainty 2.1×10^{-8}; its inferred Q is therefore not resolved. The three preferred central scales are not a common scale. For the model's *own* lepton outputs to satisfy exact Koide, the required x is $137.045540384 > x_0$. Spacelike vacuum polarization moves x downward, so no $Q > 0$ achieves exact Koide by this substitution.



3. The energy scales supplied by the manuscript itself

The prescribed reference radius is $r_e = \alpha\hbar/(m_e c)$, hence

$$Q_r \equiv \frac{\hbar c}{r_e} = \frac{m_e c^2}{\alpha} = 70.0253 \text{ MeV}, \qquad E_* \equiv \frac{3\hbar c}{2r_e} = 105.0379 \text{ MeV}. \tag{3}$$

Substitution of the spacelike $x_{\rm eff}(Q)$ alone gives the following *diagnostic*, not a consistent full RG improvement:

Q	$x_{\rm eff}(Q)$	R_μ	$m_\tau c^2$ [MeV]	R_p
Thomson limit	137.0359992	206.7682845	1776.95694	1836.1526734
$\hbar c/r_e$	136.1597542	205.4539170	1765.53905	1824.3233662
E_*	136.0635681	205.3096379	1764.28570	1823.0248544
Target	—	206.7682827	1776.93000	1836.1526734

The physically distinguished momenta of the model therefore *worsen* all three numerical agreements if only the electromagnetic coupling is replaced.

4. Running masses in a common $\overline{\rm MS}$ scheme

Use one-loop electromagnetic beta and mass anomalous dimensions, with $S(\mu) = \sum_{f\,\text{active}} N_c Q_f^2$:

$$\frac{d\alpha}{d\ln\mu} = \frac{2S\alpha^2}{3\pi}, \qquad \frac{d\ln\bar{m}_\ell}{d\ln\mu} = -\frac{3\alpha}{2\pi} \quad (\ell = e, \mu, \tau). \tag{4}$$

For each lepton, one-loop pole–$\overline{\rm MS}$ matching at its own mass and piecewise integration are used:

$$\frac{m_\ell^{\rm pole}}{\bar{m}_\ell(\mu)} = 1 + \frac{\alpha(\mu)}{\pi}\left(1 + \frac{3}{4}\ln\frac{\mu^2}{(m_\ell^{\rm pole})^2}\right) + O(\alpha^2). \tag{5}$$

The weakly constrained low-energy hadronic contribution is represented by one *effective* common light-quark threshold, fixed to reproduce the PDG 2022 $\overline{\rm MS}$ electromagnetic charge $\hat{\alpha}^{-1}(m_\tau) = 133.471$. The resulting effective threshold is 0.1209 GeV; it is a computational surrogate, not a constituent or current quark mass. Above this threshold, charm and bottom thresholds are included. Pure one-loop electromagnetic running above a few GeV omits electroweak matching and higher-order effects; it is sufficient to check whether order-unity changes in the required inverse coupling are avoided.

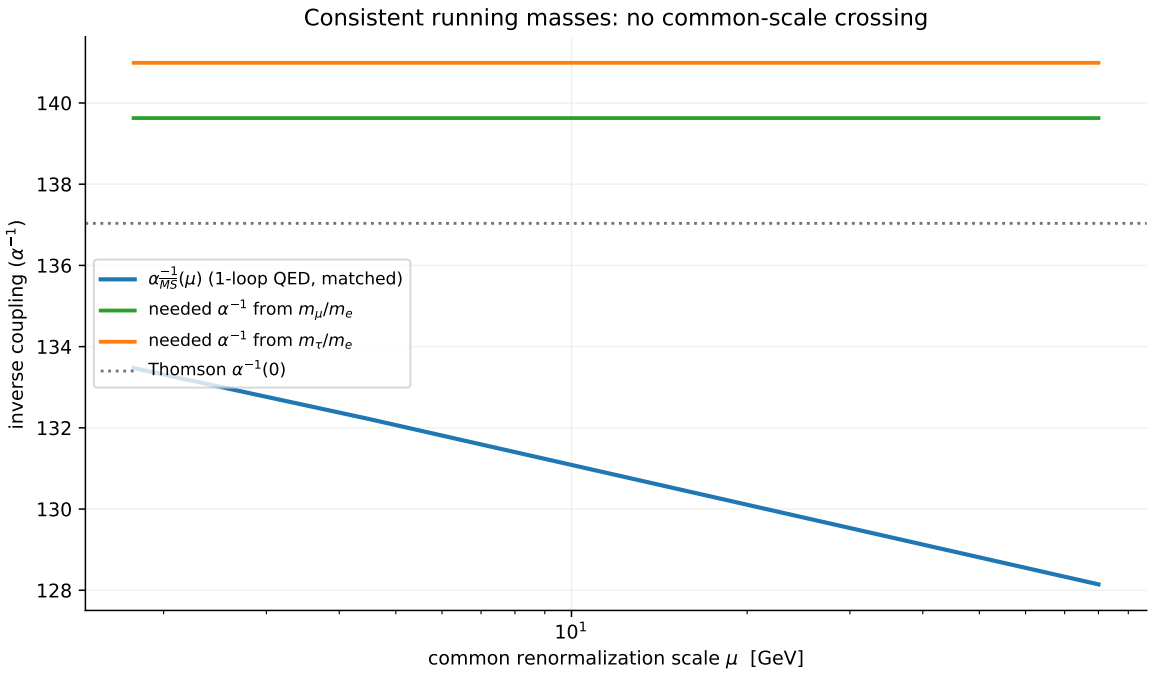
At $\mu = m_\tau$, the result is:

$$R_\mu^{\overline{\rm MS}} = 210.657394, \qquad R_\tau^{\overline{\rm MS}} = 3578.226230, \qquad x_{\rm EM}^{\overline{\rm MS}} = 133.471. \tag{6}$$

To satisfy the manuscript's unchanged equations, these ratios would instead demand

$$x_\mu^{\rm needed} = \frac{2}{3}(R_\mu^{\overline{\rm MS}} - 17/14) = 139.628739, \quad x_\tau^{\rm needed} = \frac{2}{51}(R_\tau^{\overline{\rm MS}} + 17) = 140.989264. \tag{7}$$

Neither agrees with 133.471, nor with each other. Since at one loop all three leptons have the same electromagnetic mass anomalous dimension, their $\overline{\rm MS}$ mass ratios are constant above m_τ, whereas $\alpha(\mu)$ keeps running. Thus the mismatch grows above m_τ; there is no crossing in the region scanned. Small lepton-Yukawa effects at electroweak scales do not alter this conclusion.

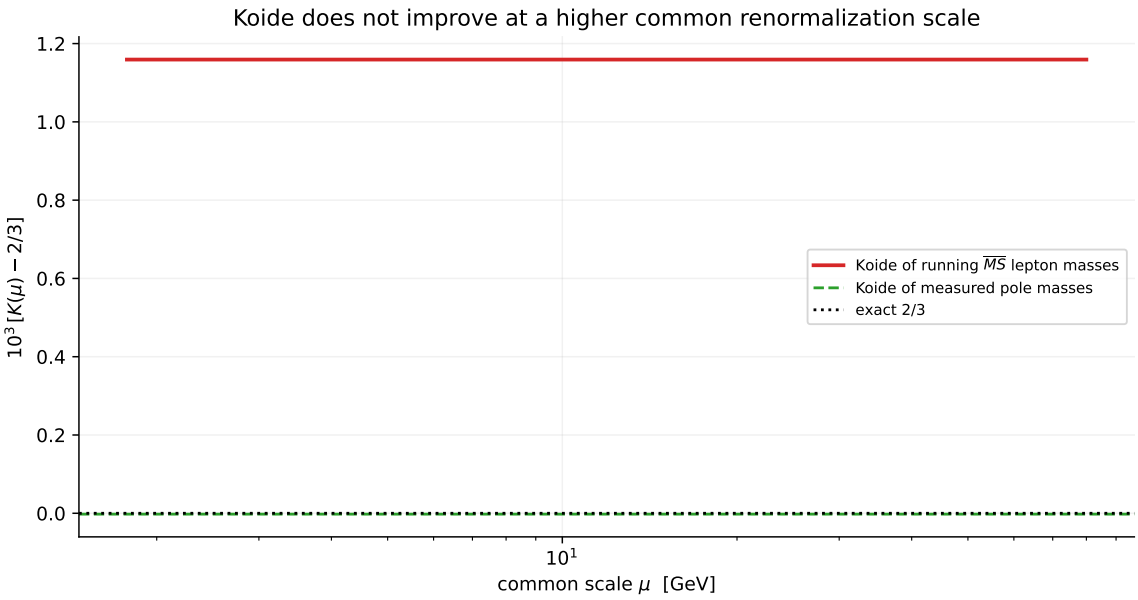


5. Koide under RG

Using the tau central value above, the observed pole-mass Koide combination is

$$K^{\rm pole} = 0.6666644634, \qquad K^{\overline{\rm MS}}_{\text{one-loop, } \mu \geq m_\tau} = 0.6678258296, \tag{8}$$

compared with $2/3 = 0.6666666667$. Its proximity to $2/3$ is a property of the pole-mass boundary data, not of the corresponding common-scale $\overline{\rm MS}$ masses. At one-loop QED, the ratios are exactly scale-invariant above the tau threshold; consequently K is flat, not driven back toward $2/3$. This agrees qualitatively with Xing–Zhang (2006), who find a roughly 0.2% difference between Koide using running and pole masses at the weak scale. Differences of order 10^{-4} in the illustrative $K^{\overline{\rm MS}}$ estimate are within the scope of omitted higher-order electroweak and threshold effects; the sign and magnitude of the displacement are robust.



6. RG covariance of the proposed structural law

An equation like $R_\mu = 3/[2\alpha(\mu)] + 17/14$ cannot be RG invariant over an interval when R_μ is (approximately) invariant but $\alpha(\mu)$ runs. If one introduces an additional radius multiplier $\lambda(\mu)$, requiring fixed ratios and finite offsets forces $[\alpha(\mu)\lambda(\mu)]^{-1} = \text{constant}$, i.e. $\lambda(\mu) \propto \alpha(\mu)^{-1}$. This cancels the electromagnetic coupling in the radius $r = \lambda\alpha\hbar/(m_e c)$, changing the original H1 prescription. It is a new postulate, not an RG prediction of the paper.

The proton pole mass is RG invariant: it does not obey the perturbative current-quark mass anomalous dimension. A consistent hadronic calculation would need QCD dynamics, including $\Lambda_{\rm QCD}$, matrix elements and a specified QED/QCD separation. Replacing $\alpha(0)$ by $\alpha(Q)$ in the proton formula is therefore only a diagnostic.

Conclusion. No special nonzero energy emerges from these tests. The original mass relations are most naturally read as *Thomson-limit pole-mass phenomenology*. Ordinary electromagnetic RG flow does not improve their simultaneous common-scale realization, and Koide becomes less accurate for running masses.

References. Kritov, *Clifford Geometry and Mass Relations*, v13 (supplied manuscript). PDG 2022, *Electroweak Model and Constraints on New Physics*, section 10.2.2, https://pdgaws.lbl.gov/2022/reviews/rpp2022-rev-standard-mod pdf. Z.-z. Xing and H. Zhang, *Phys. Lett. B* **635** (2006) 107, https://doi.org/10.1016/j.physletb.2006.02. 051. S. Antusch, K. Hinze, S. Saad, *Phys. Rev. D* **113** (2026) 095011, https://doi.org/10.1103/fdcc-ycph.